PROJECT 04 REPORT

Smart Temperature and Humidity Monitor using DHT11

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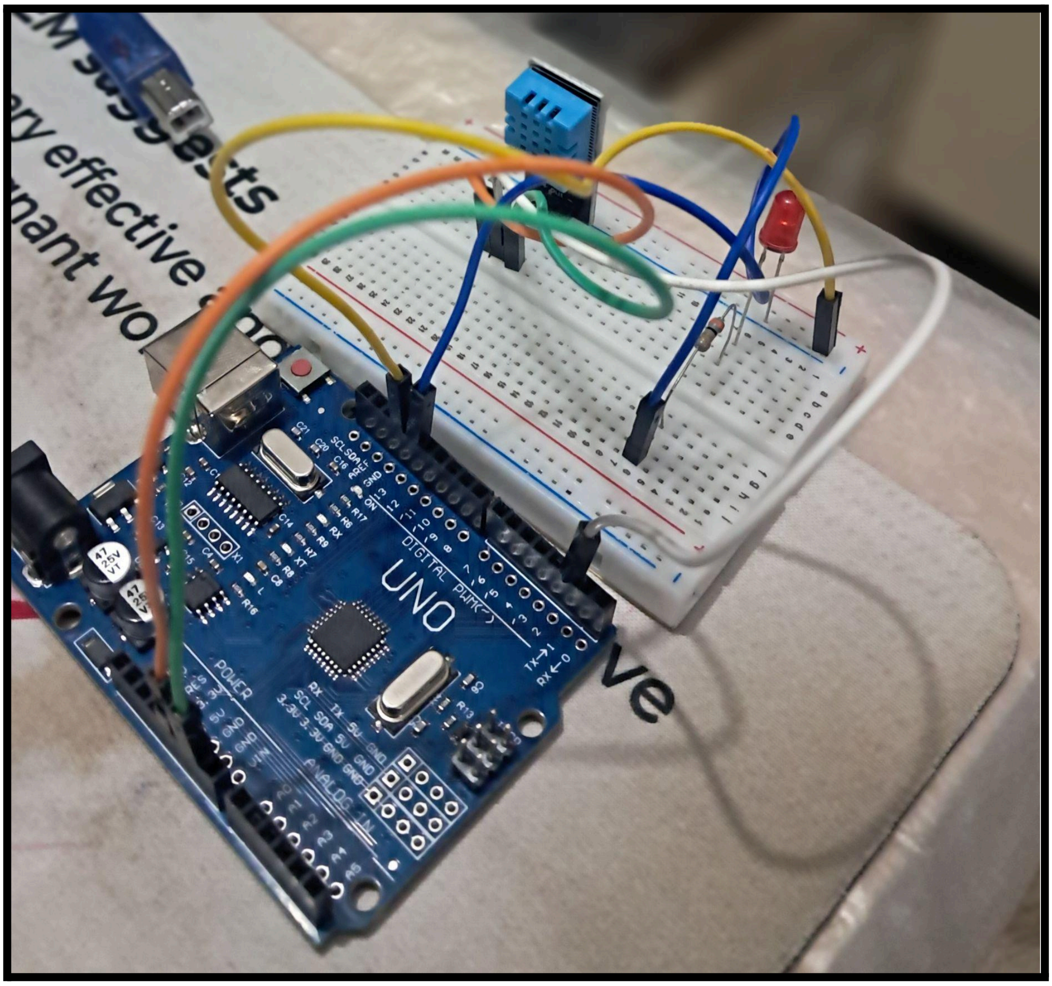
Duration : 1 day



1. Objective

The objective of this project was to develop an environmental monitoring system using the DHT11 sensor and Arduino Uno and introduction of digital sensors, external libraries and object creation.

The system measures real time temperature and humidity and displays the readings through the Serial Monitor.

2. Components Used

Component	Quantity
Arduino Uno R3	1
Breadboard	1
DHT11 Temperature & Humidity Sensor	1
LEDs (Green)	1
Resistors(300Ω)	1
Jumper Wires	Multiple
USB Cable	1

3. Software Used

- Arduino IDE
- Serial Monitor
- DHT Sensor Library (Adafruit)
- Adafruit Unified Sensor Library
- Windows Laptop

4. Theory

The DHT11 is a digital sensor which measures both temperature and humidity. Unlike analog sensors, the DHT11 internally processes sensor data and transmits the measured values digitally to the Arduino.

The sensor provides:

- **Temperature → °C**
- **Humidity → %**

The DHT11 communicates with Arduino using a digital data line and requires an external library to decode and access the sensor readings.

Temperature and humidity values can be read using:

- **dht.readTemperature();**
- **dht.readHumidity();**

These values can be used for monitoring, alert generation and environmental automation systems.

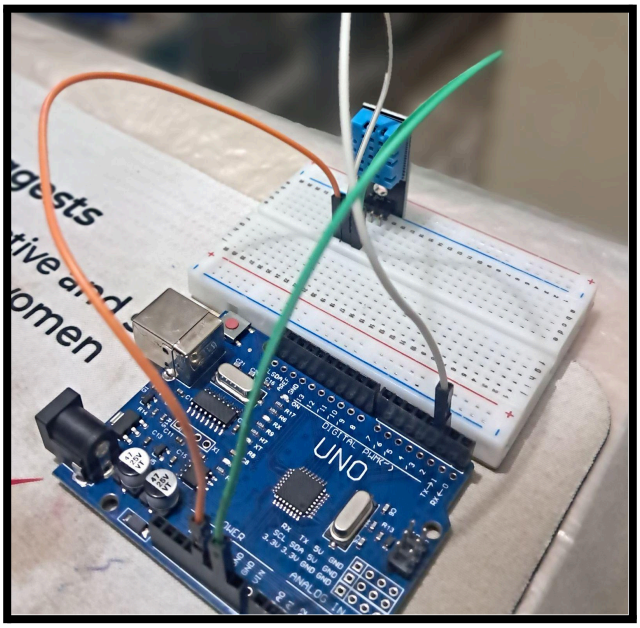
5. Tasks Performed

➢ Understanding DHT11 Sensor

The working principle of the DHT11 sensor was studied. The sensor's pin configurations, operating mechanism and digital communication method were explored.

The concepts of temperature measurement, humidity measurement, external libraries, classes and objects were studied before hardware implementation.

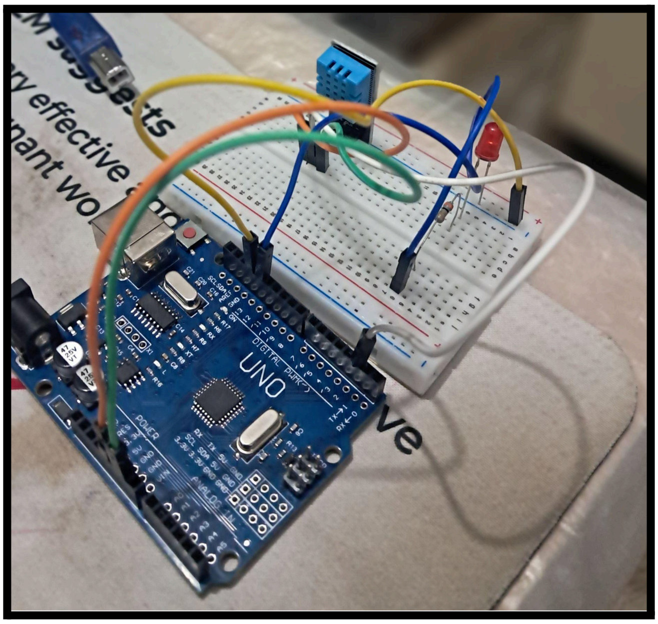
➢ Reading Temperature and Humidity Values



Video

The DHT11 sensor was connected with Arduino and real time temperature and humidity readings were obtained using the DHT library and displayed on Serial Monitor under room conditions.

➢ Temperature Alert System



Video

A temperature based alert system was implemented using the DHT11 sensor and an LED indicator. The measured temperature was continuously compared with a predefined threshold value.

6. Learning Outcomes

During this project, the following concepts were learned:

- DHT11 Sensor Operation
- Temperature Measurement
- Humidity Measurement
- Digital Sensors
- External Arduino Libraries
- Object Creation and Library Usage
- Sensor Initialization
- Environmental Monitoring

- Threshold Based Decision Making
- RealTime Data Acquisition
- Serial Communication
- Sensor Based Alert Systems

7. Problems Faced

I. Understanding the difference between analog sensors and digital sensors.
II. Selecting an appropriate temperature threshold for the alert system.**(Multiple readings were recorded over time and the threshold values were adjusted experimentally based on observed environmental conditions)**
III. LCD integration could not be completed.**(The LCD and I2C interface module were separate components and were not attached together so LCD integration was postponed as a future enhancement)**

8. Observations

- Temperature readings varied slightly over time.
- Humidity remained relatively stable during testing.
- DHT11 provides direct temperature and humidity values without requiring ADC conversion.
- External libraries simplify communication with digital sensors.
- Proper threshold selection is important as environmental conditions directly affect sensor readings.

9. Applications and Future Scope

Temperature and humidity monitoring systems are widely used in weather stations, smart homes, greenhouses, storage facilities, laboratories and industrial environments.

The project can be upgraded further by using LCD based display,IoT monitoring using ESP32,Mobile app integration,Data logging and analysis.

10. Conclusion

The Smart Temperature and Humidity Monitor project introduced the use of digital sensors for real world environmental monitoring applications.

This project serves as an important step toward more advanced embedded and IoT based monitoring systems involving displays, wireless communication, digital sensors,cloud connectivity and intelligent automation.